

# SAI MADHUSUDAN GUNDA

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## Education

**International Institute of Information Technology, Hyderabad (IIIT-H)**

**2024 – Present**

*M.S. by Research in Computer Science and Engineering*

*CGPA: 9.0/10*

**International Institute of Information Technology, Hyderabad (IIIT-H)**

**2020 – 2024**

*B.Tech in Computer Science and Engineering*

*CGPA: 8.4/10*

## Experience

**BharatGen – National AI Mission**

**Jan 2025 – Present**

*Generative AI Researcher*

*IIIT Hyderabad*

- Developed **Patram-7B-Instruct**, a multilingual vision-language model for document understanding, OCR, layout parsing, question answering, classification, and summarization across **Indian languages**.
- Built scalable data pipelines for synthetic QA generation, key-value extraction, and summarization, producing large-scale multimodal training samples for instruction tuning and supervised fine-tuning.
- Implemented efficient multi-node distributed training framework using **DeepSpeed ZeRO-3** on **256 H100 GPUs**.

**Center for Visual Information Technology (CVIT), IIIT Hyderabad**

**May 2023 – Present**

*Research Student – Computer Vision*

*Advisor: Prof. Ravi Kiran Sarvadevabhatla*

- Built ML systems for grounded document question answering, historical manuscript OCR, and multimodal document understanding, with a focus on scalable training, evaluation, and inference.
- Published **2 research papers** in top computer vision venues including CVPR and WACV, spanning document grounding, OCR, and multilingual tasks.

**Language Technologies Research Centre (LTRC), IIIT Hyderabad**

**Jan 2024 – Apr 2024**

*Undergraduate Researcher – Natural Language Processing*

*Advisor: Prof. Radhika Mamidi*

- Developed classification models for Indian languages, achieving **92%** accuracy on multilingual datasets by fine-tuning.
- Conducted experiments on cross-lingual transfer and low-resource language classification, contributing to improved understanding of multilingual NLP for Indic languages.

## Publications

**CURIO: Curvature-Aligned and Efficient OCR for Low-Resource Historical Manuscripts**

**WACV 2026**

[Project Page](#)

Proposes a curvature-aware OCR model that handles warped and degraded historical Indic manuscripts, improving recognition accuracy for low-resource scripts.

**M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA**

**CVPR 2026**

[Project Page](#)

Introduces a mask-based grounding framework that localizes multi-span evidence at phrase, line, and block levels for document question answering.

**UniLipi: A Unified Multi-Script OCR for Historical Indic Manuscripts**

**ICDAR 2026**

[Project Page](#)

Unified OCR model across 13 Indic scripts with script-aware synthetic data, achieving 6.9% CER and strong cross-script generalization.

**Multi Script OCR System for Handwritten Indic Manuscripts**

**DAS 2026**

[Project Page](#)

Demo of Unified OCR model across Indic scripts.

**DoCoG: Mask-based Multi-Type Grounded Chain-of-Thought for Document QA (Oral)**

**ECCV 2026**

[Project Page](#)

Grounded document question answering with step-wise reasoning .

**DocLayout-VL: Hierarchical, Open-set, and Promptable Document Layout Segmentation**

**ECCV 2026**

[Project Page](#)

Promptable foundational document layout model achieving SOTA results across document layout benchmarks.

## Projects

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### Scalable Vision-Language Document Inference with vLLM | *Python, PyTorch, vLLM*

- Built high-throughput inference pipelines for document models using **vLLM**, enabling efficient processing of **2M+ documents**.
- Reduced inference latency by **2.5×** through prompt batching, cached image embeddings, and dynamic request scheduling.
- Benchmarked throughput, memory footprint, and response quality across different configurations for document QA workloads.

### Multi-Page Document Grounding with Agentic RAG | *Python, PyTorch, RAG, Vision-Language Models*

- Extended **M3Grounder** to a multi-page retrieval-augmented generation pipeline, enabling grounded QA over long documents with evidence spanning multiple pages.
- Built a page-level retriever and grounding-aware answerer that uses fine-grained evidence localization to generate answers tied to supporting regions.

### Document QA and Key-Value Extraction Pipeline | *Python, PyTorch, OCR, Transformers*

- Designed an end-to-end pipeline for document question answering and key-value extraction using OCR, layout-aware features, and multimodal transformers.
- Integrated synthetic data generation, grounding, and evaluation workflows to support scalable experimentation on document intelligence tasks.

### Mixture-of-Experts Routing for Document Models | *PyTorch, Distributed Inference*

- Explored routing strategies to direct document inputs to specialized experts, achieving **15%** compute savings with less than **1%** accuracy drop.
- Evaluated accuracy-latency-compute trade-offs to identify efficient inference strategies for multimodal models.

### Recommendation Engine with Collaborative Filtering | *Python, Scikit-learn, SQL, Flask*

- Developed a hybrid recommendation system combining collaborative filtering and content-based methods on the **MovieLens 25M** dataset and served predictions via a REST API.

### Real-Time Object Detection & Tracking Pipeline | *Python, PyTorch, OpenCV*

- Built an end-to-end detection and tracking system using **YOLOv8** for multi-object tracking on surveillance videos, with ONNX export for optimized deployment.

## Technical Skills

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**Programming:** Python, C/C++, JavaScript, SQL, HTML/CSS

**Deep Learning:** PyTorch, Hugging Face Transformers, DeepSpeed, vLLM, Ms Swift, Unsloth

**Tools:** Docker, Git, Linux, AWS (S3, EC2), MongoDB, MERN Stack

**Core Areas:** Computer Vision, Document AI, NLP, Distributed Training, Model Optimization

## Achievements

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- Awarded **Dean's List** at IIIT Hyderabad for two consecutive years (2021, 2022).
- Secured **1st Place** in **Megathon** Hackathon conducted by IIIT Hyderabad and Qualcomm.
- Achieved **7th position** in **Amazon ML Challenge** (2024).
- National-level exam ranks: UGEE (19), JEE Mains (5055), JEE Advanced (3240 AIR).